DO YOU KNOW?

Elasticity is predominantly used to assess the change in consumer demand as a result of a change in price.

Relationship among Marginal revenue [MR], Average revenue [AR] and Elasticity of demand[η_d].

We know that $R(x)=px$

i.e., $R = px$

and $\eta_d = -\frac{p}{x}\cdot\frac{dx}{dp}$

Now, MR $= \frac{d}{dx}(R)$

$= \frac{d}{dx}(px)$

$= p + x\frac{dp}{dx}$

$= p\left[1+\frac{x}{p}\cdot\frac{dp}{dx}\right]$

$= p\left[1+\frac{1}{\frac{p}{x}\cdot\frac{dx}{dp}}\right]$

$= p\left[1-\frac{1}{-\frac{p}{x}\cdot\frac{dx}{dp}}\right]$

$= p\left[1-\frac{1}{\eta_d}\right]$

i.e. $MR = AR\left[1-\frac{1}{\eta_d}\right]$ (or)

$\eta_d = \frac{AR}{AR-MR}$

Example 6.1

The total cost function for the production of x units of an item is given by $C(x)=\frac{1}{3}x^3+4x^2-25x+7$.

Find

(i) Average cost function

(ii) Average variable cost function

(iii) Average fixed cost function

(iv) Marginal cost function and

(v) Marginal Average cost function

Solution:

$C(x)=\frac{1}{3}x^3+4x^2-25x+7$

(i) Average cost:

$AC=\frac{C}{x}=\frac{1}{3}x^2+4x-25+\frac{7}{x}$

(ii) Average variable cost:

$AVC=\frac{f(x)}{x}=\frac{1}{3}x^2+4x-25$

(iii) Average fixed cost:

$AFC=\frac{k}{x}=\frac{7}{x}$

(iv) Marginal cost:

$MC=\frac{dC}{dx}$ (or) $\frac{d}{dx}(C(x))$

$=\frac{d}{dx}\left[\frac{1}{3}x^3+4x^2-25x+7\right]$

$=x^2+8x-25$

(v) Marginal Average cost:

$MAC=\frac{d}{dx}[AC]=\frac{d}{dx}\left[\frac{1}{3}x^2+4x-25+\frac{7}{x}\right]$

$=\frac{2}{3}x+4-\frac{7}{x^2}$

Example 6.2

The total cost C in Rupees of making x units of a product is $C(x)=50+4x+3\sqrt{x}$. Find the marginal cost of the product at 9 units of output.

Solution:

$C(x)=50+4x+3\sqrt{x}$

Marginal cost $(MC)=\frac{dC}{dx}=\frac{d}{dx}[C(x)]$

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